

$^{26}\text{Mg}(^{18}\text{O}, 2\alpha n\gamma)$ **2021Go09**

2021Go09: $^{26}\text{Mg}+^{18}\text{O}\rightarrow^{44}\text{Ca}\rightarrow 2\alpha+n+^{35}\text{S}$ fusion-evaporation reaction. Exp 1 determined the beam energy for producing high-spin states of ^{35}S : an 80-MeV ^{18}O beam was delivered from the JAEA tandem accelerator facility and impinged on a 0.5-mg/cm² thick, self-supporting metallic enriched foil of ^{26}Mg . γ rays were detected using the GEMINI-II array of 14 Ge detectors with BGO Compton suppressor shields. The reaction channel was selected by detecting the evaporated charged particles with a 4 π silicon detector. Measured E_γ , I_γ , $\alpha\gamma\gamma$ -coin, $\gamma\gamma(\theta)$ (ADO). Exp 2 used the same ^{18}O beam energy, the same target, and the same charged-particle detectors as Exp 1. The beam was delivered from the ALTO-Tandem accelerator facility at IPN Orsay. γ rays were detected using the ORGAM array of 13 Ge detectors based on the EUROGAM with the BGO scintillator for the anti-Compton suppressor. Compared with large-scale shell-model calculations with the SDPF-MSD4 interaction.

 ^{35}S Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
0.0 [#]	3/2 ⁺		
1990.6 [#] 4	7/2 ⁻		
3594.2 [#] 7	(7/2 ⁺)		
3815.0 6	(9/2 ⁻)		
4023.1 5	(11/2 ⁻)		
4822.4 6	(9/2 ⁺)		
4899.6 7	(9/2 ⁺)		
5009.8 6	(11/2 ⁻)		
5412.4 6	(9/2 ⁺)		
5878.0 [#] 6	(11/2 ⁺)		
7179.6 [#] 7	(15/2 ⁺)		
8023.9 [#] 9	(17/2 ⁺)		
8755.7 [#] 13	(17/2)	<1 ps	$T_{1/2}$: estimated by 2021Go09 from residual Doppler energy shifts observed for the 732 γ and 1576 γ peaks.
10048.4 [#] 10	(19/2)		
12469.1 [#] 15	(21/2)		

[†] From a least-squares fit to γ -ray energies.

[‡] As given in **2021Go09**, which quotes the J^π assignments for levels below 8.5 MeV from **2014Ay01** but adds parentheses.

2021Go09 assigns J^π for levels above 8.5 MeV based on the measured $\gamma\gamma(\theta)$ (ADO) ratios.

[#] Seq.(A): γ cascade based on the g.s.

 $\gamma(^{35}\text{S})$

$R_{\text{ADO}}^{\gamma 1} = I_{\gamma 1}(47^\circ \text{ gated by } \gamma_2 \text{ at all}) / I_{\gamma 1}(86^\circ \text{ gated by } \gamma_2 \text{ at all})$. Expected values are ≈ 0.8 for dipole ($\Delta J=1$) and ≈ 1.0 for stretched quadrupole ($\Delta J=2$) transitions.

E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [†]	Comments
465.5 2	14 1	5878.0	(11/2 ⁺)	5412.4	(9/2 ⁺)	M1	$R_{\text{ADO}}=0.67$ 11.
731.6 15	8 3	8755.7	(17/2)	8023.9	(17/2 ⁺)	D	$R_{\text{ADO}}=0.43$ 9.
844.3 6	20 3	8023.9	(17/2 ⁺)	7179.6	(15/2 ⁺)	M1	$R_{\text{ADO}}=0.51$ 8.
868.2 4	20 2	5878.0	(11/2 ⁺)	5009.8	(11/2 ⁻)	E1	$R_{\text{ADO}}=0.54$ 21.
978.4 10	26 7	5878.0	(11/2 ⁺)	4899.6	(9/2 ⁺)	M1	
986.6 3	26 1	5009.8	(11/2 ⁻)	4023.1	(11/2 ⁻)	M1	$R_{\text{ADO}}=0.91$ 13.
1007.2 11	4 2	4822.4	(9/2 ⁺)	3815.0	(9/2 ⁻)	E1	$R_{\text{ADO}}=0.87$ 15.
1055.6 4	39 10	5878.0	(11/2 ⁺)	4822.4	(9/2 ⁺)	M1	$R_{\text{ADO}}=0.67$ 18.
1228.3 6	11 2	4822.4	(9/2 ⁺)	3594.2	(7/2 ⁺)	M1	
1301.6 3	56 11	7179.6	(15/2 ⁺)	5878.0	(11/2 ⁺)	E2	$R_{\text{ADO}}=1.40$ 17.

Continued on next page (footnotes at end of table)

$^{26}\text{Mg}(^{18}\text{O}, 2\alpha n\gamma)$ 2021Go09 (continued) $\gamma(^{35}\text{S})$ (continued)

E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [†]	Comments
1305.3 4	5 3	4899.6	(9/2 ⁺)	3594.2	(7/2 ⁺)	M1	
1389.0 8	9 2	5412.4	(9/2 ⁺)	4023.1	(11/2 ⁻)	E1	
1576.3 14	14 3	8755.7	(17/2)	7179.6	(15/2 ⁺)	D	
1824.4 6	48 2	3815.0	(9/2 ⁻)	1990.6	7/2 ⁻	M1	
1854.5 42	28 14	5878.0	(11/2 ⁺)	4023.1	(11/2 ⁻)	E1	
1990.5 4	100	1990.6	7/2 ⁻	0.0	3/2 ⁺	M2+E3	$R_{\text{ADO}}=1.30$ 8.
2032.4 4	70 16	4023.1	(11/2 ⁻)	1990.6	7/2 ⁻	E2	
2063.3 14	32 6	5878.0	(11/2 ⁺)	3815.0	(9/2 ⁻)	E1	$R_{\text{ADO}}=0.58$ 41.
2283.5 10	14 3	5878.0	(11/2 ⁺)	3594.2	(7/2 ⁺)	E2	$R_{\text{ADO}}=0.83$ 17.
2420.6 11	21 6	12469.1	(21/2)	10048.4	(19/2)	D	$R_{\text{ADO}}=0.50$ 42.
2868.7 8	26 2	10048.4	(19/2)	7179.6	(15/2 ⁺)	Q	$R_{\text{ADO}}=0.98$ 13.
3594.4 11	35 2	3594.2	(7/2 ⁺)	0.0	3/2 ⁺	E2	$R_{\text{ADO}}=0.93$ 20.

[†] From $\gamma(\theta)$ and $\gamma\gamma(\theta)(\text{ADO})$ data in 2021Go09.

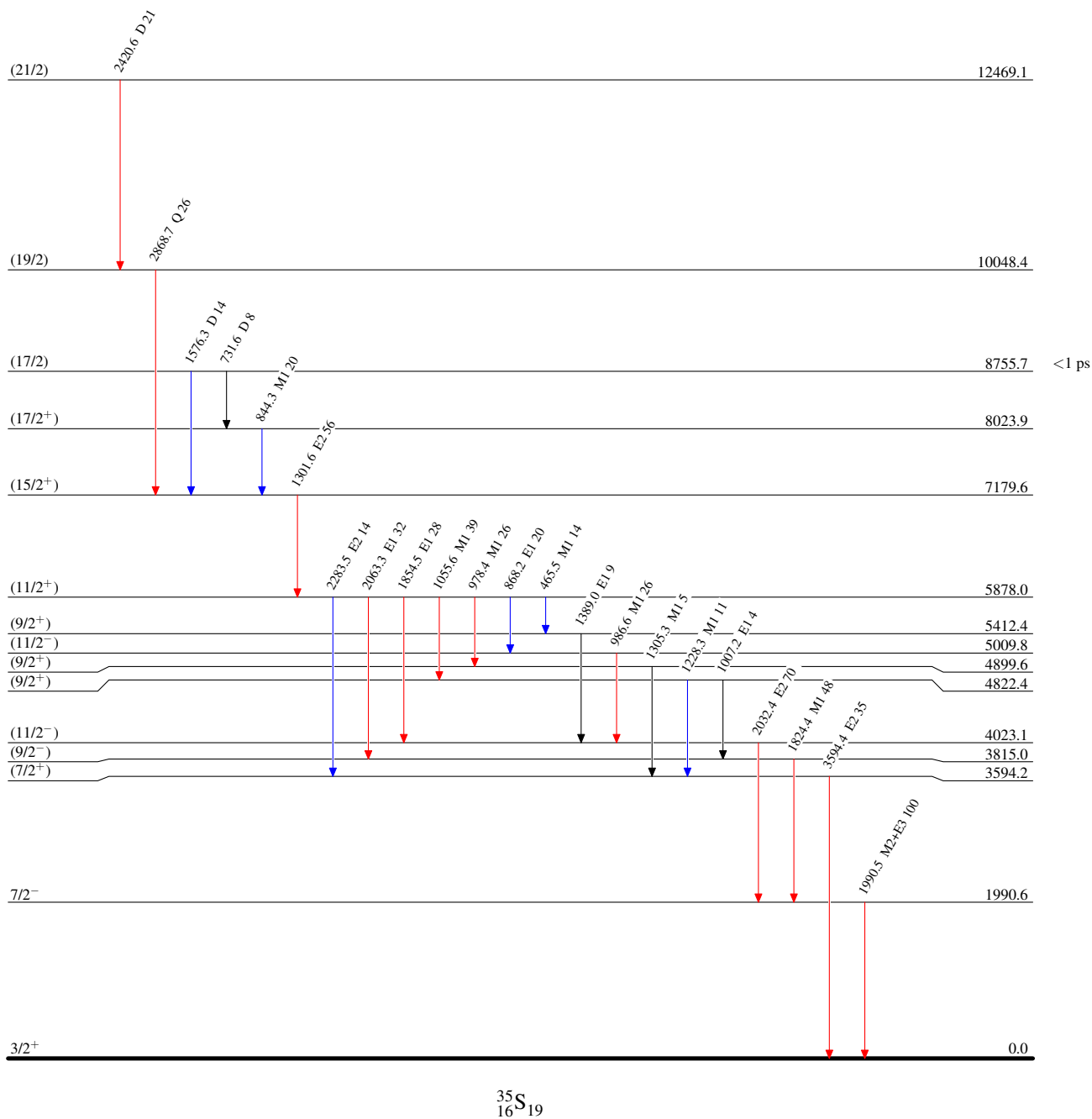
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Level Scheme

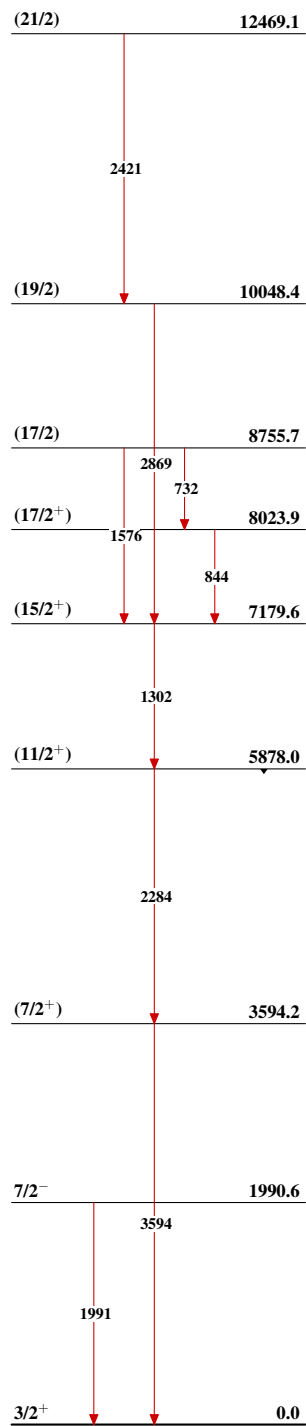
Intensities: Relative I_γ

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$



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 Seq.(A): γ cascade based on the
g.s

 $^{35}_{16}\text{S}_{19}$